IEEE Access
Multidisciplinary : Rapid Review : Open Access Journal

Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.2026.DOI

A Physiologically Interpretable Multimodal Model for Joint Sleep Staging and Respiratory-Event Detection in Subacute Ischemic Stroke

MD WAHIDUZZAMAN SUVA[1], ESM-E MOULA CHOWDHURY ABHA[1], MD IMTIAJ ALAM SAJIN[1], AND MD IFTEKHARUL MOBIN[1]

[1]Department of Computer Science and Engineering, American International University-Bangladesh, Dhaka 1229, Bangladesh

Corresponding author: Md Iftekharul Mobin (e-mail: iftekhar.mobin@aiub.edu).

The authors received no specific funding for this work.

ABSTRACT A clinical polysomnogram records far more than the brain: alongside the electroencephalogram it captures eye movements, muscle tone, the electrocardiogram, nasal and oral airflow, breathing effort, and blood oxygen saturation. In patients recovering from ischemic stroke, a group in which sleep-disordered breathing is common, these cardiorespiratory signals carry the disease itself. We build one model that reads the full polysomnogram and produces two clinical outputs at once: the sleep stage of every 30-second epoch and a per-epoch label for respiratory events. The model encodes the neural and cardiorespiratory signals in two parallel streams, fuses them, follows the night with a bidirectional recurrent decoder, and drives a staging head and a respiratory head, with a direct connection carrying the cardiorespiratory signal to the respiratory head so oxygen desaturation reaches the decision at full strength. We benchmark it against deep-learning and classical baselines on iSLEEPS, the first public polysomnography corpus of subacute ischemic stroke, under patient-independent ten-fold cross-validation. The single model stages sleep at 0.739 accuracy with Cohen's κ of 0.634, at the ceiling this cohort imposes on every model family, and detects respiratory events at 0.782 area under the ROC curve — an epoch-level, screening-grade read-out rather than event-by-event scoring — a capability no staging model possesses. Evaluated zero-shot on two independent public corpora with no dataset-specific tuning, the same weights transfer without degradation, indicating a representation that is physiological rather than specific to one centre. Code, extracted features and trained checkpoints are released.

INDEX TERMS Cardiorespiratory signals, feature fusion, ischemic stroke, multi-task learning, multimodal learning, polysomnography, sleep apnea, sleep staging.

I. INTRODUCTION

SLEEP is scored in 30-second epochs into five stages, Wake (W), N1, N2, N3, and rapid-eye-movement (REM), following the American Academy of Sleep Medicine (AASM) rules [1]. The overwhelming majority of automated methods read a single electroencephalogram (EEG) channel and ask one question: how accurately can the five stages be classified? Two decades of work on that question have moved from convolutional and recurrent encoders [2]–[4] through attention and sequence models [5]–[7] to large self-supervised EEG foundation models [8]–[11]. On healthy sleepers these methods reach roughly 85% accuracy, and recent reviews argue the field is near a ceiling set by inter-scorer disagreement [12].

A clinical polysomnogram (PSG), however, is a recording of the whole sleeping body. It carries the EEG, eye movements, muscle tone, the electrocardiogram (ECG), nasal and oral airflow, thoracic and abdominal breathing effort, and peripheral oxygen saturation (SpO_2). These cardiorespiratory channels are where sleep-disordered breathing writes itself: a paused breath, a chest that keeps straining against a closed airway, a fall in blood oxygen, and the arousal that rescues the breath. A model that reads only the EEG cannot see any of this, and so cannot report on the disorder that most damages a patient's sleep. We use the full polysomnogram, and we ask a question that the staging race does not: what does a model gain, and on which clinical task, when it reads the whole recording rather than the brain alone.

We study this on iSLEEPS [13], released in 2026 as the first public PSG corpus dedicated to subacute ischemic stroke. The cohort is well matched to the question. Sleep-disordered breathing is common in the weeks after a stroke and is tied to worse recovery [14], [15], and in this corpus the median patient carries a scored respiratory event in 13% of epochs, with 77 of 96 patients above 5% (Fig. 1). The cardiorespiratory signal is therefore abundant and clinically meaningful in exactly the population where breathing matters most. The published staging baselines on this corpus reach 74.7% accuracy for an LSTM, about ten points below the healthy state of the art, and a concurrent study reports that deep networks attend to non-physiological regions on this cohort [16], so the setting also stresses every staging architecture we can bring to it.

We present a two-stream, multi-task model that treats staging and respiratory-event detection as one joint problem, and we place it inside a broad benchmark on the same folds. Our contributions are:

- **C1. One model for two clinical read-outs from the full polysomnogram.** It stages sleep at 0.739 accuracy (κ 0.634) and, from the same forward pass, detects respiratory events at 0.782 AUC, with a direct cardiorespiratory path to the respiratory head so oxygen-desaturation cues reach the decision undiminished (Section V).
- **C2. Healthy-sleep deep models fail on the injured brain.** Deep networks trained from scratch, transferred from healthy sleep, or applied to the raw multichannel signal collapse to 0.61–0.69 accuracy on this cohort (Table 3), well below the published single-channel baseline; the compact feature-based model reaches 0.739 and, uniquely, also produces a respiratory read-out. Staging accuracy here reflects a cohort ceiling; the joint respiratory output is what we contribute.
- **C3. A physiologically grounded respiratory read-out.** Removing the airflow channel from which the events were scored leaves detection at 0.778 AUC, because the decision rests on respiratory effort above all, together with oxygen desaturation, cardiac variability and cortical arousals, encoded redundantly enough that no single channel is indispensable (Section VI-G).
- **C4. Causal, per-modality attribution.** A leave-one-out modality-ablation grid shows each channel carries the task its physiology governs: removing the cardiorespiratory stream lowers only respiratory detection (Wilcoxon $p = 0.004$), while removing the ocular channel lowers only staging, giving a falsifiable and physiologically interpretable account of the model (Table 5).

Together, these results reframe the problem on this cohort: the useful gains come from reading the whole sleeping body and attributing each output to the physiology that produces it; deeper single-channel staging does not help on this cohort. The remainder of the paper develops the model (Section V) and tests each claim in turn.

II. RELATED WORK

A. SINGLE-CHANNEL DEEP SLEEP STAGING

Automated staging matured on one EEG channel. DeepSleepNet paired convolutional feature extraction with a bidirectional LSTM [2]; SeqSleepNet framed the night as sequence-to-sequence labeling [3]; AttnSleep introduced attention over multi-resolution convolutions [5]; TinySleepNet and U-Time reduced the parameter budget while holding accuracy [4], [7]; and XSleepNet fused raw and spectral views [6]. More recent backbones push this further: SleepTransformer adds interpretability and uncertainty to an attention model [17], L-SeqSleepNet models whole-night cycles [18], ZleepAnlystNet and FlexSleepTransformer explore separated training and flexible channel sets [19], [20], and linear-time state-space backbones reach competitive accuracy [21], [22]. These models set the accuracy on healthy corpora but depend on the cortical micro-events that stroke disturbs, which is why they transfer poorly to patients [16].

B. MULTI-CHANNEL AND FOUNDATION MODELS

Later work reads several PSG channels. U-Sleep stages from arbitrary EEG and EOG derivations across many cohorts [23], and self-supervised EEG foundation models such as LaBraM, CBraMod, EEGPT, and BIOT learn transferable representations from large unlabeled corpora [8]–[11]. SleepFM extends this idea to several PSG modalities [24], while cross-scale foundation models and standardized benchmarks continue to appear [25]–[27], self-supervised pretraining is now systematically evaluated for label efficiency [28], transfer and domain adaptation improve robustness across cohorts [29], and recent surveys and clinical benchmarks catalogue the field [30], [31]. The eye and muscle channels help most with REM and Wake, which a single EEG channel resolves poorly. Reported gains from adding respiratory channels to *staging* are small and cohort-dependent; our benchmark quantifies this directly on a high-burden clinical cohort and separates the staging effect from the respiratory one.

C. DETECTION OF SLEEP-DISORDERED BREATHING

A parallel line of work detects apnea and hypopnea from breathing signals. Airflow and effort shape, oximetry, and ECG-derived heart-rate variability are all informative, and oxygen desaturation is a defining marker of an event [32]. These detectors usually run at fine time resolution on the raw breathing waveforms and are trained on that task alone. We instead detect respiratory events on the staging epoch grid, jointly with staging, from a shared representation, so that one model produces both clinical outputs and the two tasks support each other during training.

D. SLEEP AND BREATHING AFTER STROKE

Sleep is both disturbed by stroke and predictive of recovery, and sleep-disordered breathing is common in the subacute phase and linked to poorer functional outcome [14], [15], [33]. Focal injury also changes the micro-events that scoring relies on, the spindles, slow waves, and muscle atonia [34]–[38]; quantitative EEG markers such as the brain symmetry index further track hemispheric injury [39]. A model that reports on breathing as well as stage is therefore of direct clinical value in this population. Sleep staging on ischemic stroke itself is nascent, reported only as baselines on the iSLEEPS corpus [13], and no prior method addresses staging and respiratory detection jointly, from the full polysomnogram, on a stroke population; we make this comparison quantitative in Table 13.

III. CRITICAL GAPS AND LIMITATIONS OF PRIOR WORK

Three gaps run through the literature above, and each maps onto one of our contributions. *(i) Single-task, single-signal design.* Almost every staging model reads one or a few EEG channels and predicts only the stage, while apnea detectors read breathing signals and predict only events; the two literatures rarely meet in one model, leaving the clinician with two separate tools. Contribution C1 answers this with a single model that reads the full polysomnogram and produces both outputs. *(ii) Evaluation on healthy sleepers.* Staging accuracy is reported almost entirely on healthy corpora such as Sleep-EDF [40], [41], where deep and foundation models reach a ceiling, while performance on focal brain injury is largely unmeasured. Contribution C2 benchmarks strong deep architectures, classical gradient-boosting pipelines [42]–[44], and graph models [45] on a subacute-stroke cohort, and shows that the healthy-built networks collapse. *(iii) Attribution, reliability, and cost.* Interpretability is usually post-hoc and unfalsifiable, predictive uncertainty is seldom modeled [46], and the compute and energy footprint of heavy models is rarely reported [47]. Contributions C3 and C4 answer this with a physiologically grounded, non-circular respiratory read-out and a causal modality-ablation that attributes each output to the channel its physiology governs, in a compact model.

IV. DATASET AND PREPROCESSING

iSLEEPS [13] contains overnight PSG from 100 subacute ischemic-stroke patients (mean age 50.5; 77 male, 23 female), scored by experts into the five AASM stages. We exclude one recording after an audit: the data records of SN28 are byte-identical to those of SN15, a duplication consistent with the anonymization and re-numbering the dataset authors describe, and keeping it would place a training subject into the test fold in nine of ten fold assignments, giving $N = 99$. Sleep staging uses all 99 patients; the respiratory task uses the $N = 96$ with complete cardiorespiratory channels, and we state which N applies to each reported number. The corpus totals 89,532 epochs. The stage distribution (Table 1) is strongly imbalanced, with N2 dominant and N1 and N3 scarce, which is the clinical norm.

TABLE 1. Stage distribution and respiratory-event prevalence in the corpus we model ($N = 99$).

Stage	W	N1	N2	N3	R
Share (%)	26.6	10.2	42.3	8.9	12.1

Respiratory events: 16.0% of epochs positive; median 13% per patient (range 0 to 60.5%); 77 of 96 patients above 5%.

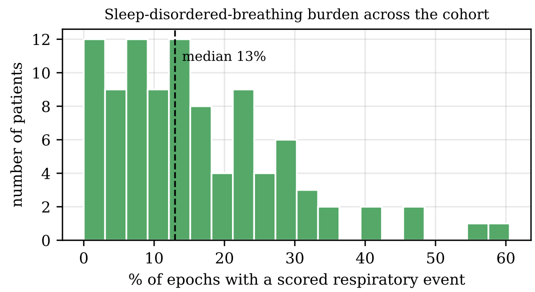

FIGURE 1. Sleep-disordered-breathing burden across the cohort. Each bar counts patients by the fraction of epochs that carry a scored respiratory event. The median patient sits at 13% and many exceed 30%, so the cardiorespiratory signal is abundant.

a: Channels we read, and channels we do not

A raw recording holds 22 traces. We read the 14 that carry sleep-stage or breathing information. The *neural stream* takes the four EEG derivations (C4:M1, C3:M2, O2:M1, O1:M2), the two EOG channels (E1:M2, E2:M2), and the submental EMG. The *cardiorespiratory stream* takes the ECG, nasal and oral airflow, thoracic effort, abdominal effort, the summed effort trace, $\mathrm{SpO_2}$, and pulse. We leave out eight traces for stated reasons: periodic limb movement records a different disorder; snore is scored from the same airway sound as the respiratory events and would make the second task circular; plethysmography is already summarized by the $\mathrm{SpO_2}$ and pulse channels we keep; body position is coarse and slow; and the ambient-light and battery traces are device housekeeping, not physiology. All neural channels are resampled to 100 Hz and the cardiorespiratory channels to a common 25 Hz grid; $\mathrm{SpO_2}$, recorded natively at 4 Hz, is upsampled to this grid, so its effective resolution remains that of the 4 Hz oximeter. Signals are read in physical units, which matters because $\mathrm{SpO_2}$ near 90%, pulse near 78 bpm, and microvolt EEG live on very different scales. Per-epoch respiratory labels are the corpus's scored events mapped onto the 30 s grid, and an epoch is positive when it overlaps an event.

V. PROPOSED METHOD

A. PROBLEM FORMULATION

A night is a sequence of T epochs. For epoch t we form a neural feature vector $\mathbf{f}_t^{\mathrm{eeg}} \in \mathbb{R}^{188}$ and a cardiorespiratory feature vector $\mathbf{f}_t^{\mathrm{car}} \in \mathbb{R}^{14}$, and we learn a subject-independent map

$$f_\theta : \left(\mathbf{f}_{1:T}^{\mathrm{eeg}}, \mathbf{f}_{1:T}^{\mathrm{car}}\right) \longmapsto \left(\hat{y}_{1:T}^{\mathrm{stg}}, \hat{y}_{1:T}^{\mathrm{apn}}\right), \tag{1}$$

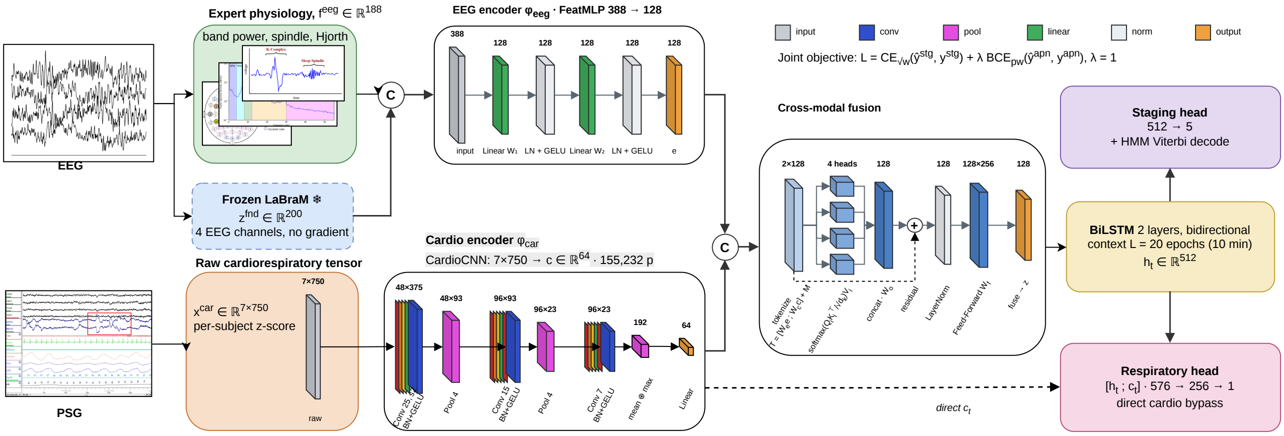

FIGURE 2. The two-stream, multi-task model. Both encoders are shown with their internals expanded: layer shapes above each block, operation below. Block height encodes units for the multilayer perceptron and temporal length for the convolutional stack. Cross-modal fusion is detailed separately (lower right). Dashed outlines mark frozen parameters; colours are keyed at the foot of the figure.

where $\hat{y}_t^{\text{stg}} \in \Delta^4$ is a distribution over the five stages and $\hat{y}_t^{\text{apn}} \in [0, 1]$ is the probability that epoch t contains a respiratory event (Fig. 2).

B. PHYSIOLOGICAL FEATURE REPRESENTATIONS

a: Neural features (188)

Per channel we take absolute and relative band powers from the Welch spectrum $P(f)$ in delta (0.5–4), theta (4–8), alpha (8–12), sigma (12–16), and beta (16–30) Hz; the relative power in band b is

$$\mathrm{RP}_b = \frac{\int_{f\in b} P(f)\, df}{\int_{0.5}^{30} P(f)\, df}. \tag{2}$$

We add spectral entropy $H = -\sum_i p_i \log p_i$ with p_i the normalized spectrum, the 95% spectral-edge frequency, the three Hjorth descriptors [48], and time-domain statistics. Sleep spindles are detected on the sigma-band analytic envelope $e(t) = |\mathcal{H}\{x_\sigma\}(t)|$ from the Hilbert transform $\mathcal{H}$, with density counted above a per-epoch threshold; slow-wave amplitude, ocular-movement energy on the EOG, and tonic EMG level complete the set.

b: Cardiorespiratory features (14)

From the seven cardiorespiratory channels we compute SpO_2 mean, minimum, variability, and desaturation depth (median minus minimum); pulse mean and variability; ECG amplitude and line-length; airflow amplitude and line-length; thoracic, abdominal, and summed effort amplitudes; and thoraco-abdominal asynchrony, the loss of coordination between chest and abdomen that marks an obstructed breath,

$$\mathrm{asyn}_t = \left|\, \mathrm{corr}(x_t^{\text{tho}},\ x_t^{\text{abd}}) \,\right|. \tag{3}$$

Both feature sets are standardized per patient, which removes person-to-person offsets so the model generalizes across people.

C. TWO-STREAM ENCODERS

Each stream is a two-layer perceptron with LayerNorm,

$$e_t = \mathrm{GELU}\big(\mathrm{LN}(W_2\, \mathrm{GELU}(\mathrm{LN}(W_1 \mathbf{f}_t^{\text{eeg}})))\big) \in \mathbb{R}^{128}, \tag{4}$$

$$c_t = \mathrm{GELU}\big(\mathrm{LN}(U_2\, \mathrm{GELU}(\mathrm{LN}(U_1 \mathbf{f}_t^{\text{car}})))\big) \in \mathbb{R}^{64}. \tag{5}$$

LayerNorm rather than BatchNorm is used because the recurrent windows are short and mix patients within a batch, and the cardiorespiratory stream is deliberately narrower because 14 slow descriptors carry far less structure than 188 neural features.

D. CROSS-MODAL FUSION

The two embeddings become two tokens with learned modality-type embeddings $M \in \mathbb{R}^{2\times D}$, $D = 128$,

$$T = \big[\, W_e\, e_t \ ;\ W_c\, c_t \,\big] + M \ \in\ \mathbb{R}^{2\times D}. \tag{6}$$

Four-head scaled dot-product attention lets each modality read the other,

$$\begin{aligned} \mathrm{head}_i &= \mathrm{softmax}\!\left(\frac{(TW_i^Q)(TW_i^K)^\top}{\sqrt{d_k}}\right) TW_i^V, \\ A &= \big[\mathrm{head}_1; \ldots; \mathrm{head}_4\big] W^O, \end{aligned} \tag{7}$$

followed by a residual connection, normalization, and a projection of the two refined tokens,

$$\tilde{T} = \mathrm{LN}(T + A), \qquad z_t = \mathrm{GELU}\big(W_f\, \mathrm{vec}(\tilde{T})\big) \in \mathbb{R}^D. \tag{8}$$

We also evaluate a plain concatenation in place of attention; the two perform alike (Section VI-F), so the reported gain reflects the added modality; the fusion design contributes little.

E. TEMPORAL DECODER AND TWO HEADS

Sleep is sequential and respiratory events cluster, so the fused vectors of an $L = 20$ epoch window (10 minutes) are read by a two-layer bidirectional LSTM. Each direction runs the standard recurrence

$$\begin{aligned} i_t &= \sigma(W_i[z_t, h_{t-1}]), & f_t &= \sigma(W_f[z_t, h_{t-1}]), \\ o_t &= \sigma(W_o[z_t, h_{t-1}]), & g_t &= \tanh(W_g[z_t, h_{t-1}]), \\ s_t &= f_t \odot s_{t-1} + i_t \odot g_t, & h_t &= o_t \odot \tanh(s_t), \end{aligned} \quad (9)$$

and the forward and backward states are concatenated into $h_t \in \mathbb{R}^{256}$. The staging head is a linear map $\hat{y}_t^{\text{stg}} = \mathrm{softmax}(W_s h_t)$. The respiratory head reads h_t together with a direct copy of the per-epoch cardiorespiratory embedding c_t,

$$\hat{y}_t^{\text{apn}} = \sigma\big(W_2\, \mathrm{GELU}(W_1[h_t; c_t])\big). \quad (10)$$

This direct connection matters: without it the respiratory decision must survive a fusion trained mostly for staging, and the fall in oxygen that defines an event is weakened along the way. The full model holds 0.86 million parameters.

F. TRAINING AND DECODING

The two tasks are trained together with equal weight, $\mathcal{L} = \mathcal{L}^{\text{stg}} + \lambda\,\mathcal{L}^{\text{apn}}$ with $\lambda = 1$, combining a class-weighted staging cross-entropy and a positive-weighted respiratory cross-entropy,

$$\mathcal{L}^{\text{stg}} = -\sum_t \sum_k w_k\, y_{t,k}^{\text{stg}} \log \hat{y}_{t,k}^{\text{stg}}, \quad (11)$$

$$\mathcal{L}^{\text{apn}} = -\sum_t \big[\pi\, a_t \log \hat{y}_t^{\text{apn}} + (1 - a_t)\log(1 - \hat{y}_t^{\text{apn}})\big]. \quad (12)$$

The staging weights use the square root of inverse class frequency, $w_k \propto \sqrt{N/(Kn_k)}$ for $K = 5$ classes with n_k examples, which corrects the scarce N1 and N3 stages without giving up overall accuracy, and the respiratory term is positive-weighted by $\pi = n_-/n_+$ to match the 16% event rate. At inference the staging posteriors are smoothed by a hidden-Markov Viterbi pass whose transition matrix A^{H} and prior are counted from the training labels [49],

$$\delta_t(j) = \max_i \big[\delta_{t-1}(i) + \log A_{ij}^{\mathrm{H}}\big] + \log \hat{y}_t^{\text{stg}}(j), \quad (13)$$

recovered by back-tracking, which restores the natural persistence of sleep. The respiratory head is left unsmoothed. Optimization uses AdamW with cosine annealing and early stopping on a held-out set of patients.

VI. EXPERIMENTS

A. PROTOCOL AND METRICS

All results use ten-fold, patient-independent cross-validation, so no patient appears in both training and test, and within each training split a disjoint set of patients is held out for early stopping. Staging is scored by accuracy, macro-F1 $\frac{1}{K}\sum_k \mathrm{F1}_k$, and Cohen's κ. The respiratory task is scored by the area under the ROC curve (AUC) and average precision (AP), which suits the 16% event rate.

Every number we report is a mean over the ten folds with its standard deviation, and the folds are fixed across all configurations so that comparisons are paired. Because a single training seed can flatter a result, we retrained the headline configuration under five seeds on the same folds. Staging accuracy varies by 0.003 across seeds against 0.032 across folds, and respiratory AUC by 0.004 against 0.035 — roughly a tenth as much. The uncertainty in these results is therefore dominated by which patients fall in which fold, which is a property of a cohort this size, and not by the optimiser's starting point. We report the seed-42 run throughout for consistency with the released artifacts, and note that its respiratory AUC sits at the top of the five-seed range (seed mean 0.705), so the reader should treat 0.782 as the optimistic end of a 0.701–0.711 interval rather than as a point estimate.

TABLE 2. Training configuration, shared across all folds, re-implemented baselines, and ablations.

Setting	Value
Framework / hardware	PyTorch, NVIDIA RTX 2060 (6 GB)
Optimizer	AdamW
Learning rate	1×10^{-3}
Weight decay	1×10^{-4}
LR schedule	cosine annealing
Sequence length L	20 epochs (10 min)
Batch size	32
Max epochs / patience	45 / 8
Gradient clipping	5.0
Staging class weights	$\surd$ inverse-frequency
Respiratory pos. weight	$\pi = n_-/n_+$
Loss balance λ	1.0
Staging post-processing	HMM Viterbi smoothing
Cross-validation	10-fold, patient-independent
Random seed	42

B. IMPLEMENTATION AND HYPERPARAMETERS

The model is implemented in PyTorch and trained on a single NVIDIA RTX 2060 GPU (6 GB) under Windows, with all randomness seeded to 42 so that runs are deterministic and reproducible. Table 2 lists the training configuration, which is identical across all folds, all baselines we re-implement, and every ablation, so that reported differences come from the model and not from tuning.

C. COMPARED METHODS

We compare the model against deep networks evaluated on the same folds: convolutional and recurrent networks trained from scratch (DeepSleepNet, AttnSleep, a four-channel CNN–BiLSTM), a network transferred from Sleep-EDF, and a raw-signal convolutional network on all 14 channels, alongside the published single-channel baselines for the corpus [13]. All numbers are ten-fold patient-independent, at the hidden-Markov operating point where a model provides one.

D. STAGING BENCHMARK AND COMPARISON

Table 3 shows a clear pattern: every deep network, whether trained from scratch, transferred from healthy sleep, or applied to the raw multichannel signal, stays between 0.61

TABLE 3. Sleep staging on iSLEEPS (ten-fold, patient-independent): the proposed model against deep networks trained from scratch, transferred from healthy sleep, or applied to the raw multichannel signal. "feat" marks engineered features; best per column in bold.

	Model	Input	Params ↓	Acc ↑	Macro-F1 ↑	κ ↑
Deep networks	CNN-ResNet18 [13]	1 EEG	∼11 M	0.617	0.544	0.480
	DeepSleepNet [2]	1 EEG	∼21 M	0.615	0.567	0.483
	AttnSleep [5]	1 EEG	∼0.5 M	0.686	0.629	0.577
	CNN + BiLSTM	4 EEG	∼0.6 M	0.613	0.565	0.485
	Sleep-EDF transfer	4 EEG	∼0.6 M	0.623	0.582	0.488
	Raw multimodal CNN	14 ch	0.95 M	0.655	0.612	0.531
Proposed feature-based multi-task model	MM-Net (neural only)	7 ch feat	0.77 M	0.721	0.635	0.603
	MM-Net (neural + cardio)	14 ch feat	0.86 M	0.739	**0.670**	**0.634**

and 0.69 accuracy, 15 to 20 points below the same architectures' published accuracy on healthy sleepers. The compact, feature-based model reaches 0.739, above every deep baseline, which matches the concurrent finding that depth alone does not help on this cohort [16]: with under a hundred patients a convolutional network cannot rediscover what decades of sleep science already encode in the features. Staging accuracy is nonetheless capped: the published single-channel LSTM for this corpus reaches 0.747, and no method we are aware of exceeds 0.75 on it. Section VI-K shows why — the staging learning curve has flattened, so the ceiling is a property of this cohort rather than of any one model. We therefore do not position the model as a staging leader. What sets it apart is a second output: it is the only model that also detects respiratory events. A representative held-out night is shown in Figure 3: the predicted hypnogram follows the reference through the night's cycles, and the stage-probability ribbon exposes where the model is confident and where it hesitates.

E. PER-CLASS STAGING PERFORMANCE

Table 4 breaks staging down by stage. The pattern is the one this cohort imposes on every method: N2, Wake, and REM are recovered well, N3 is moderate, and N1 is the hard stage, because it is transient and rare. The raw convolutional network reaches the highest N1 F1, which fits its sensitivity to short events, while the proposed model leads on the stable stages. The row-normalised confusion matrix (Fig. 4) shows where the errors land: N1 is absorbed mainly into Wake and N2, and part of N3 is read as N2. Comparing the neural-only and full rows isolates what the cardiorespiratory stream contributes to staging, and the answer is narrow: N1 improves by 0.019 (Wilcoxon $p = 0.045$ over thirty fold-values) and no other stage moves ($p \geq 0.15$). That the gain lands on N1 is consistent with its physiology — it is the transitional stage, the hardest to score, and the one most often coincident with a respiratory event.

F. MODALITY-ABLATION GRID

This ablation is also our attribution method. Because the model is feature-based and never sees a raw spectrogram, a gradient heat-map or Grad-CAM would have no faithful spatial support; we therefore attribute each output by *causal* in-

TABLE 4. Per-class staging F1 (Wake, N1, N2, N3, REM), ten-fold patient-independent, at the hidden-Markov operating point. The neural-only and full rows are means over the same thirty fold-values. Best per column in bold.

Model	W ↑	N1 ↑	N2 ↑	N3 ↑	R ↑
Raw multimodal CNN	0.74	**0.37**	0.73	0.56	0.67
MM-Net (neural only)	**0.79**	0.28	0.80	0.71	**0.76**
MM-Net (neural + cardio)	0.79	0.30	**0.80**	**0.71**	0.75

tervention — deleting a physiological channel and measuring the change — which is a falsifiable test of what each modality contributes rather than a post-hoc visualization. The proposed model stages sleep at 0.739 accuracy (κ 0.634) and detects respiratory events at 0.782 AUC (0.419 average precision). To test the paper's central claim, that each modality carries the task its physiology governs, we remove each modality in turn and keep the rest, on the same folds, with three seeds giving thirty fold-values per row (Table 5). The separation between streams is clean, but within the cardiorespiratory stream the pattern is not the one the engineered features gave.

Removing the whole cardiorespiratory stream drops respiratory AUC from 0.782 to 0.654 — the largest single effect in this study — and leaves staging untouched (0.741, n.s.). Removing the neural stream does the reverse: staging falls from 0.739 to 0.680 while respiratory detection holds. No neural ablation moves respiratory AUC and no cardiorespiratory ablation moves staging, so each stream serves the outcome its physiology governs.

Within the cardiorespiratory stream, however, only respiratory effort is individually necessary (−0.038 AUC, Holm $p < 0.001$). Removing SpO_2, pulse/HRV, ECG or airflow individually costs at most 0.005 AUC and none survives correction; SpO_2, which the engineered-feature configuration ranked first, is here the most expendable channel (−0.002, $p = 0.61$). The five individual removals sum to −0.046 AUC against −0.128 for removing the stream as a whole — a factor of 2.8. The learned encoder therefore distributes respiratory information redundantly across channels: each is individually dispensable because the others overlap it, while the ensemble is essential. That redundancy is clinically favourable rather than a weakness, since twelve of the hundred recordings are missing at least one cardiorespiratory channel, and it is

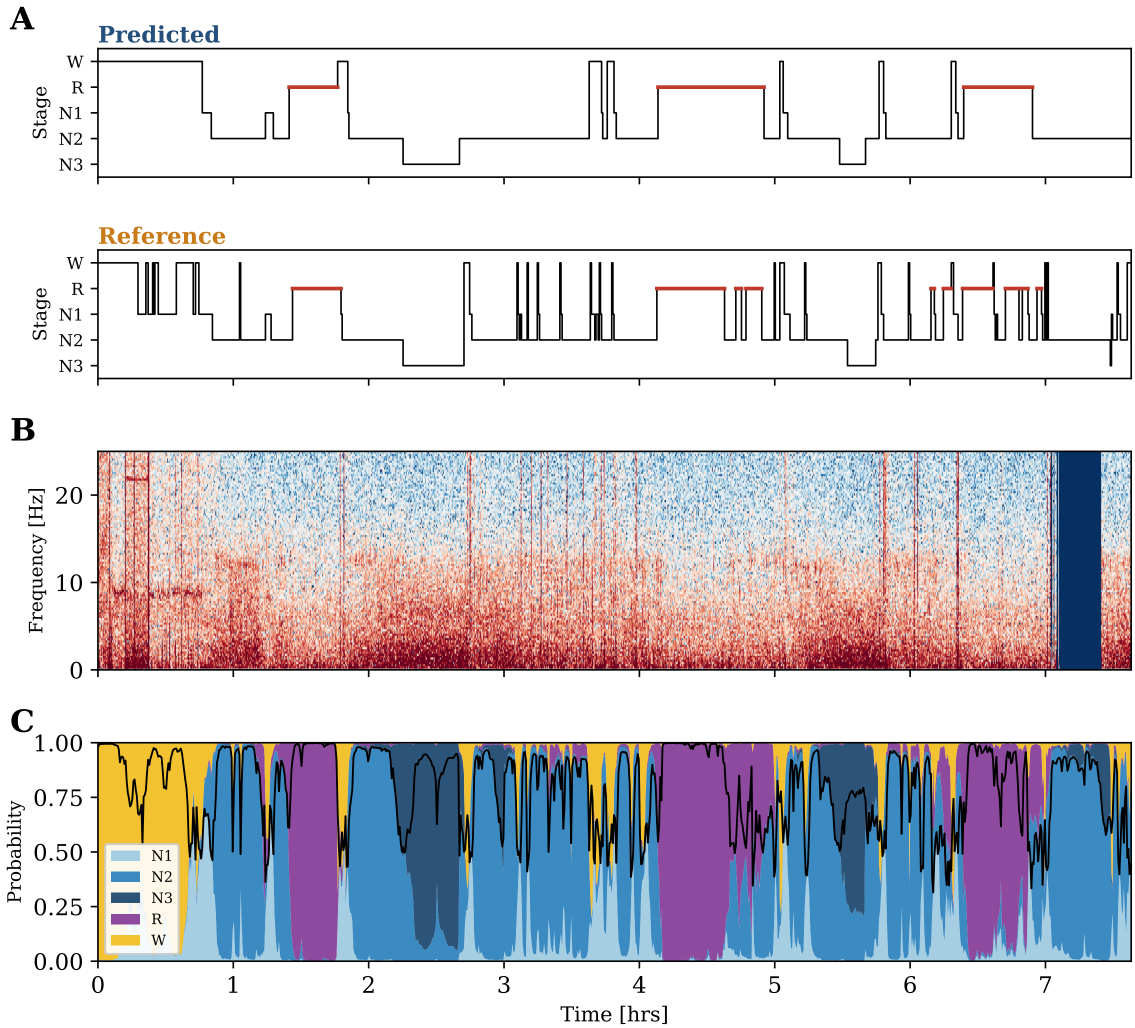

FIGURE 3. A representative held-out night (patient SN90, staging accuracy 0.80). (A) Predicted and reference hypnograms, REM highlighted. (B) EEG spectrogram, C4 derivation, 0–25 Hz. (C) Per-epoch stage-probability ribbon; the black line is the maximum posterior.

physiologically coherent — the effort belts register the paradoxical thoraco-abdominal motion of an obstructive event as it happens, whereas desaturation is a lagged consequence of it. Attention and plain concatenation stage and detect alike, so the gain comes from the added modality, and the fusion mechanism has little effect.

G. PHYSIOLOGICAL VALIDITY OF THE RESPIRATORY READ-OUT

Because the events were scored from airflow, a model that keyed on the airflow channel would be right for a trivial reason. Removing the airflow features leaves detection essentially unchanged, at 0.778 AUC (Table 5), within one fold standard deviation of the full 0.782. The model therefore does not lean on the labeling signal. It reads above all the straining of the chest and abdomen — the only channel whose removal costs significantly — together with the fall in blood oxygen, the beat-to-beat change in the heart, and the cortical arousal that ends the event, none of which is individually indispensable because the learned encoder encodes them redundantly. This is the useful regime, because it detects the bodily consequences of disordered breathing rather than repeating a measurement a technician has already made. It is

TABLE 5. Modality-ablation grid (leave-one-out), ten-fold patient-independent, three seeds. Each row removes one modality and keeps the rest; cells are mean $\pm$ SD over thirty fold-values. Largest effect per column in bold.

Modality removed	Staging acc ↑	Respiratory AUC ↑
none (full model)	0.739 ± 0.020	0.782 ± 0.038
EEG (neural)	$\mathbf{0.680 \pm 0.024}$	0.774 ± 0.041
EOG (ocular)	0.731 ± 0.023	0.777 ± 0.042
EMG (muscle)	0.735 ± 0.025	0.778 ± 0.041
SpO_2	0.742 ± 0.019	0.780 ± 0.041
respiratory effort	0.741 ± 0.018	0.744 ± 0.049
pulse / HRV	0.741 ± 0.020	0.785 ± 0.035
ECG	0.737 ± 0.020	0.777 ± 0.043
airflow	0.740 ± 0.021	0.778 ± 0.039
all cardiorespiratory	0.741 ± 0.017	$\mathbf{0.654 \pm 0.041}$

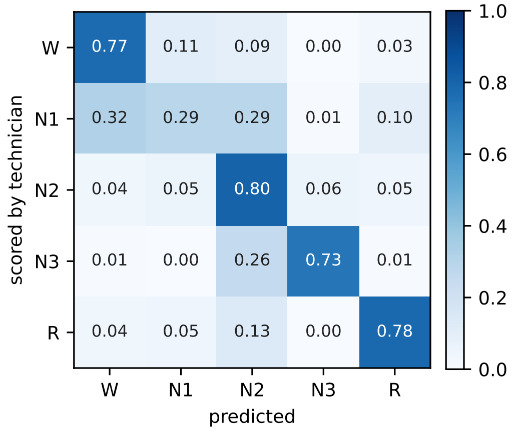

FIGURE 4. Row-normalised confusion matrix for the proposed model (pooled ten-fold test epochs, hidden-Markov operating point). Errors concentrate on N1, which is absorbed into Wake and N2.

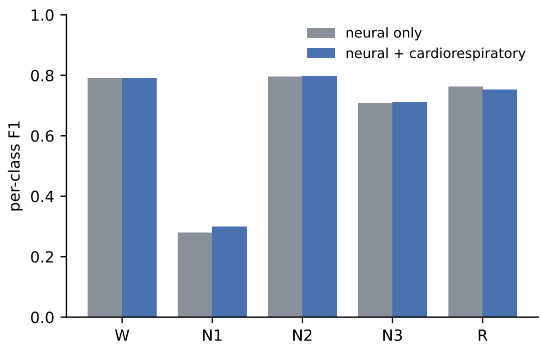

FIGURE 5. Per-class staging F1 for the neural-only and the neural-plus-cardiorespiratory model, both averaged over the same thirty fold-values.

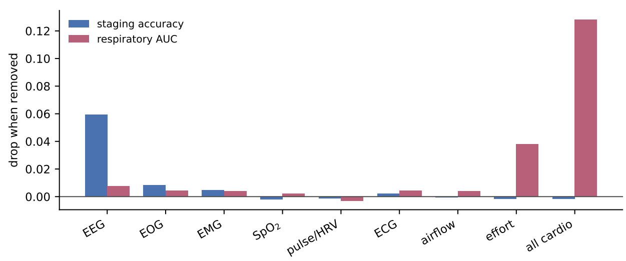

FIGURE 6. Modality-ablation grid. Drop in staging accuracy (blue) and respiratory AUC (red) when each modality is removed and the model retrained.

also the physiologically expected one: the effort belts register the paradoxical thoraco-abdominal motion of an obstructive event while it is happening, whereas desaturation follows it by several breaths.

H. RESPIRATORY BASELINES AND PER-EVENT-TYPE DETECTION

The respiratory task needs real competitors (Table 6), on the same folds and the same 14 cardiorespiratory features. A clinical desaturation rule reaches 0.596 AUC, logistic regression 0.582, and gradient boosting 0.670; a random classifier scores 0.157 average precision, the event prevalence. The proposed model reaches 0.782 AUC and 0.419 average precision, above gradient boosting but by a modest margin, so the contribution is the joint single pass and the physiological attribution above; we do not claim state-of-the-art detection accuracy. Detection also separates by event type: the model scores 0.748 AUC on hypopneas (the mild, dominant 81% of events), 0.898 on obstructive apneas, and 0.935 on central apneas. It is strongest on the most severe events, and the overall AUC is held down by the abundant hypopneas.

Scoring at the event level rather than the epoch level changes the picture, and not favourably. Grouping consecutive flagged epochs into events within each patient, a decision threshold of 0.5 detects 55% of scored events at 2.6 false alarms per hour; lowering the threshold to 0.2 raises event sensitivity to 86% but flags 68% of all epochs. More tellingly, the model recovers the wrong *number* of events: a mean predicted rate of 4.5 events per hour against a scored 11.1, and a per-patient rank correlation of only $\rho = 0.142$ ($p = 0.17$, $n = 96$) — far weaker than the $\rho = 0.491$ obtained from the aggregated epoch probability. Consecutive events separated by a few epochs merge into one flagged run, so an epoch-level detector systematically under-counts. This is why we report a burden rather than an index, and why event-level scoring is the natural next step rather than a formality.

Figure 7 shows the operating characteristics behind these numbers, computed from the pooled held-out predictions of all ten folds. The precision–recall panel is the more informative of the two: respiratory events occupy 16% of epochs, and under that imbalance a ROC curve flatters any detector. Read at fixed sensitivity, the trade-off is explicit — at 0.80 sensitivity the model holds 0.47 specificity and 0.22 precision, which is a screening signal rather than a scoring-grade detector, and we describe it as such. We also note that pooling every fold's epochs into a single curve is not the same estimator as averaging the ten per-fold scores: pooling weights folds by epoch count and mixes their score scales, giving a pooled AUC of 0.706 against the cross-validated 0.782 ± 0.034 reported in Table 6. Both are stated so the difference is visible rather than hidden; it is a fifth of one fold standard deviation.

I. CLINICAL VALIDATION AGAINST AHI AND SEVERITY

Aggregating the per-epoch respiratory probability into a per-patient burden and correlating it with the clinical apnea-hypopnea index (AHI) from the metadata gives a significant positive association (Spearman $\rho = 0.491$, $p < 0.001$, $n = 99$; Fig. 8), so the model's output tracks a clinician's summary measure rather than only an epoch-level score. Staging, in turn, degrades as sleep-disordered breathing worsens: accuracy falls from 0.770 in patients with a normal AHI to 0.708 in the severe group (Table 7). The pathology that motivates the second output also makes the first harder, which is the clinical reality of this cohort.

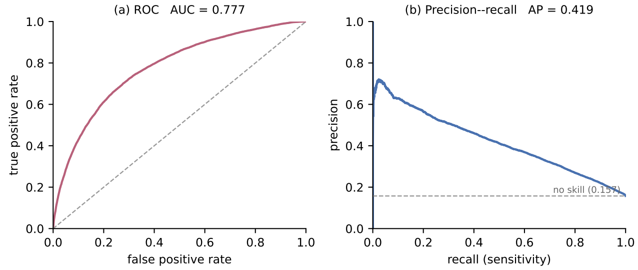

FIGURE 7. Respiratory-event detection, operating characteristics, pooled over the ten patient-independent test folds. (a) ROC; the dashed line is chance. (b) Precision–recall; the dashed line is the 0.157 event prevalence, the no-skill floor.

TABLE 6. Respiratory-event detection, ten-fold patient-independent. *(a)* Detectors trained on the 14 cardiorespiratory features on the same folds; chance-level average precision equals the event prevalence, 0.157. *(b)* AUC by event type, scored against epochs carrying no scored event, over the 96 patients with event-type annotation. Best per column in bold.

(a) Detection method (14 cardiorespiratory features)		
Method	AUC ↑	AP ↑
Desaturation rule	0.596	0.214
Logistic regression	0.582	0.193
Gradient boosting	0.670	0.290
MM-Net (proposed)	**0.782**	**0.419**

(b) MM-Net AUC by event type		
Event type	Positive epochs	AUC ↑
Hypopnea	11,605	0.748
Obstructive	1,661	0.898
Central	831	**0.935**

The association is real but modest, and it is worth saying why rather than only that it is so. Three mechanisms plausibly cap it, and each is a property of the formulation rather than of the model. First, mean per-epoch probability is a poor estimator of an hourly event rate: a 30-second epoch holding one hypopnea and one holding three both contribute a single positive, so the aggregate compresses exactly the variation the index is built from. Second, the clinical index is normalised by total sleep time, whereas our burden averages over every scored epoch including wake, and the two denominators diverge most in the patients with the most fragmented sleep — which in this cohort is the severe group. Third, the AHI itself carries substantial night-to-night variability, so a single-night reference sets a ceiling on any correlation with it. We therefore read $\rho = 0.491$ as evidence that the read-out tracks severity at the group level, and not as a claim that it estimates an index.

J. PERMUTATION IMPORTANCE: A SECOND, INDEPENDENT ATTRIBUTION

The ablation grid establishes what the model *needs*: remove a modality, retrain, and observe which head suffers. A complementary question is what a *trained* model actually uses. We

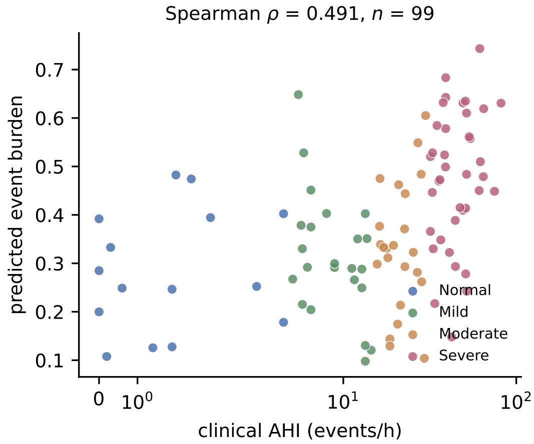

FIGURE 8. Predicted per-patient event burden (mean respiratory probability) against the clinical apnea-hypopnea index, coloured by AASM severity band. The positive association (Spearman $\rho = 0.491$, $n = 99$) shows the model's output tracks a clinician's summary measure.

TABLE 7. Sleep-staging accuracy by sleep-disordered-breathing severity (AASM AHI bands), ten-fold patient-independent; mean $\pm$ SD across patients ($N = 99$). Staging degrades monotonically as severity rises.

AHI band	AHI (events/h)	Patients	Staging acc ↑
Normal	< 5	15	0.769 ± 0.049
Mild	5–15	23	0.747 ± 0.069
Moderate	15–30	23	0.722 ± 0.088
Severe	≥ 30	38	0.719 ± 0.100

therefore permute each modality's features across epochs at inference time — the channel is still present, but no longer aligned to the epoch it describes — and measure the drop, averaged over three shuffles and the ten fold models.

The two methods can disagree, because a model may need a channel during training yet weight it lightly afterwards. On staging they do not: permuting the neural features costs 0.284 staging accuracy against 0.030 for the ocular and 0.006 for the muscle channel, the same ordering the ablation gives, and the rank correlation between the two attributions is $\rho = 0.833$ ($p = 0.010$).

On the respiratory head the agreement is narrower, and we state it precisely rather than pooling the two into a single coefficient. Both methods place respiratory effort first by a wide margin — 0.038 AUC under retraining, 0.071 under permutation, in each case well clear of the runner-up. The remaining four cardiorespiratory channels, however, have retraining effects that are not distinguishable from zero, so their rank order carries no signal and the correlation over the five is correspondingly uninformative ($\rho = 0.381$, $p = 0.35$). The claim the comparison supports is therefore the one that matters physiologically: two attributions sharing no machinery — one applied to retrained models, one to fixed ones — independently identify effort as the dominant respiratory cue. Unlike an embedding projection, that was a prediction that could have failed.

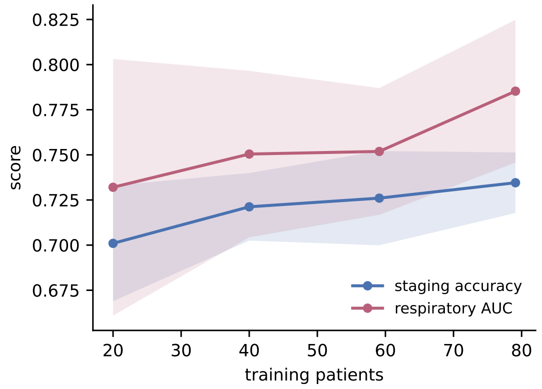

FIGURE 9. Learning curve over training-patient fraction, ten-fold patient-independent, error bars one standard deviation across folds. Staging (blue) flattens by 57 patients; respiratory detection (orange) is still climbing at 76.

K. DATA EFFICIENCY: WHICH CEILING ARE WE AGAINST?

Staging on this corpus saturates near 0.72–0.75 for every model family tried here and elsewhere. Whether that reflects the architecture or the cohort is testable: retrain the identical model on a shrinking fraction of the *training patients*, always evaluating on the full held-out fold (Fig. 9). Patients are subsampled rather than epochs, because epochs within a patient are heavily correlated.

The two heads answer differently. Staging accuracy rises $0.701 \rightarrow 0.721 \rightarrow 0.726 \rightarrow 0.739$ as training patients go from 20 to 79, with marginal gains of $+0.020$, $+0.005$ and $+0.009$, all under half a fold standard deviation and flattening. Respiratory AUC over the same range rises $0.732 \rightarrow 0.750 \rightarrow 0.752 \rightarrow 0.782$, and its *largest* increment is the final one, $+0.033$ — close to a full fold standard deviation, and steeper than any earlier step. **Staging is at or near this cohort's ceiling; respiratory detection is not, and is still accelerating.** We draw the narrower conclusion the data support: more patients of this kind would improve breathing detection and would not appreciably improve staging, so the staging ceiling is a property of the labels and the injured brain rather than of the training-set size. The final model also exceeds the feature-only configuration at every training-set size tested, so its advantage is not an artefact of the full sample.

L. HEALTHY-TRAINED MODELS APPLIED TO THE INJURED BRAIN

Table 3 retrains published architectures *on* iSLEEPS. A different and harder question is what happens when a model built for healthy sleep is applied to this cohort as published. We trained three architectures on Sleep-EDF Expanded and applied each to all 99 iSLEEPS patients zero-shot, with no adaptation of any kind (Table 8).

The result is not a degradation but a collapse. All three stage healthy sleep competently, 0.806 to 0.868 accuracy and κ 0.73 to 0.83; on stroke the same checkpoints reach 0.307 to 0.325 with κ 0.09 to 0.11, which is close to chance for a five-class problem with this prior. The failure mode is more informative than the headline: each model labels 66–78% of stroke epochs Wake against a true 26.7%, and N2 recall falls to 0.032 on the stage that is 42% of the data. These models do not lose accuracy gracefully; they stop recognising sleep architecture at all.

TABLE 8. Architectures trained on healthy sleep (Sleep-EDF Expanded) and applied zero-shot to all 99 iSLEEPS patients. "Wake called" is the fraction of stroke epochs assigned to Wake against a true 26.7%.

Model	Params	Accuracy ↑ healthy	Accuracy ↑ stroke	Wake called
CNN	0.49 M	0.806	0.325	70.5%
CNN + BiLSTM	1.15 M	0.834	0.307	65.6%
StagingSeqNet	1.08 M	**0.868**	0.308	78.4%
κ: *healthy* 0.73–0.83 → *stroke* 0.09–0.11				

The control is what makes this interpretable. A healthy-trained stager scoring 0.31 on iSLEEPS is evidence of a cohort effect only if the same checkpoint, through the same preprocessing, scores properly on healthy sleep — otherwise a broken input pipeline and a genuine domain gap produce identical numbers. Run through our pipeline on held-out Sleep-EDF recordings, the strongest checkpoint holds at 0.8682 (0.8667 ± 0.0415 per recording), so the pipeline is correct and the collapse belongs to the cohort.

Two bounds, reported together. Sleep-EDF is recorded on Fpz-Cz and Pz-Oz where iSLEEPS uses C4:M1 and O2:M1, so part of this gap is montage shift rather than pathology, and quoting the zero-shot number alone would overstate the case. Deep models trained *on* iSLEEPS with the matched montage reach 0.61–0.69 (Table 3), which puts the cost of pathology alone at roughly 0.19 against healthy-sleep performance; healthy-trained and zero-shot, with the montage shift included, the cost is 0.56. The honest claim is bounded by those two figures rather than by either alone.

M. WHAT EACH STREAM SHOULD BE REPRESENTED AS

The architecture makes two representation choices — engineered features plus a frozen self-supervised embedding on the neural side, a learned convolutional encoder over the raw signal on the cardiorespiratory side — and both were selected by experiment rather than assumed. Each family below holds every other hyperparameter fixed and varies one thing, over the same three seeds and ten patient-independent folds, with Holm correction within the family.

The cardiorespiratory features are the bottleneck, and part of the gap needs no new information. Table 9 replaces the 14 engineered cardiorespiratory features with progressively richer representations of the same underlying signal. Every step is significant, and the ordering is instructive. A *random* nonlinear expansion of the same 14 numbers recovers 0.018 AUC: that gain needs no new measurement at all, only more capacity to read what was already computed, which says

TABLE 9. **Cardiorespiratory representation, with the neural stream fixed. Ten-fold patient-independent, three seeds, thirty fold-values per row. ′ and *p* against the 14 engineered features; Wilcoxon paired on fold-values, Holm-corrected.**

Cardiorespiratory representation	Resp. AUC ↑	Staging acc	ΔAUC	Holm p
14 engineered features	0.698 ± 0.051	0.736 ± 0.014	—	—
random nonlinear expansion of the 14	0.715 ± 0.047	0.737 ± 0.017	$+0.018$	< 0.001
random linear projection of raw signal	0.731 ± 0.044	0.735 ± 0.021	$+0.034$	< 0.001
learned CNN, 0.155 M (proposed)	$\mathbf{0.782 \pm 0.039}$	0.740 ± 0.019	$+0.084$	< 0.001

TABLE 10. **Neural representation, with the cardiorespiratory stream fixed at the 14 engineered features. Ten-fold patient-independent, three seeds. ′ and *p* against the 188 engineered features, Holm-corrected within the family.**

Neural representation	Staging acc ↑	Resp. AUC	Δacc	Holm p
188 engineered features	0.741 ± 0.016	0.685 ± 0.041	—	—
frozen pretrained encoder alone	0.712 ± 0.032	0.681 ± 0.033	-0.028	< 0.001
features + pretrained (proposed)	$\mathbf{0.748 \pm 0.018}$	0.690 ± 0.037	$+0.008$	0.001
features + two pretrained encoders	0.745 ± 0.018	0.684 ± 0.043	$+0.004$	0.23

the summary features leave nonlinear structure unextracted. A random linear projection of the raw waveform adds a further 0.016, so part of what the features discard is recoverable without learning anything. The learned convolutional encoder reaches 0.782, 0.084 above the engineered features.

What this establishes is that the representation of the cardiorespiratory stream, not the detector on top of it, is what limited the respiratory read-out. We therefore claim the change of representation, not the design of the encoder that implements it.

On the neural side a pretrained encoder does not replace expert features, but does add to them. Table 10 varies the neural representation with the cardiorespiratory stream held at the engineered features. Used alone, the frozen encoder is significantly *worse* than the 188 features (-0.028, Holm $p < 0.001$): pretraining on healthy scalp EEG does not by itself substitute for physiology chosen for this task. Concatenated with the features it adds $+0.008$ accuracy (Holm $p = 0.001$), and a second pretrained encoder on top of the first adds nothing further (-0.004, $p = 0.24$).

The reading we take is narrow, because it is what the numbers support: expert physiology and self-supervised pretraining encode partly different information, so their union beats either alone; but the complementarity saturates after one pretrained encoder, so this is not a general argument for stacking foundation models.

N. CALIBRATION AND WHEN THE MODEL DECLINES TO COMMIT

A clinical reader needs to know not only how often the model is right but whether it knows when it is likely to be wrong. Two measurements answer that, both computed on held-out epochs with the raw posteriors before HMM smoothing.

The model is *overconfident*: expected calibration error is 0.076, and the gap runs one way in every confidence bin — epochs it scores above 0.9 are correct 88% of the time. Stated confidences should therefore be read as ordered, not as probabilities.

Ranking, however, is what abstention needs. Applying split-conformal prediction with the fold's validation patients as the calibration set gives empirical coverage that tracks the target closely (0.894 at a 0.90 target, 0.943 at 0.95). At $\alpha = 0.10$ the model returns a single stage for 45.8% of epochs and is correct on 86.9% of those, against 59.1% on the epochs where it returns a larger set. The abstention is therefore selecting the genuinely hard epochs rather than merely hedging, and it concentrates where a scorer would expect: N1 receives a singleton set only 19% of the time, the lowest of any stage and the one human scorers agree on least.

O. EXTERNAL VALIDATION ON TWO INDEPENDENT CORPORA

Ten-fold cross-validation is patient-independent but remains within one corpus, one centre and one acquisition protocol. We therefore trained a single model on all 96 iSLEEPS patients and evaluated it, unchanged, on two public corpora with no dataset-specific tuning: no fine-tuning, no threshold search, no re-fitted model or decision threshold (per-recording feature standardisation is applied, as in training), and hidden-Markov transitions estimated from iSLEEPS alone (Table 11).

The two corpora test different things. **Sleep-EDF Expanded** [40] is healthy sleep recorded on a frontal montage (Fpz-Cz, Pz-Oz) with no cardiorespiratory channels, so it tests the staging head across both a cohort shift and a montage shift. **ISRUC-Sleep** [50] is a sleep-disorders cohort recorded on the same central and occipital derivations we use (C4, C3, O2, O1 against the contralateral mastoid), with SpO_2, airflow, effort and ECG, and with obstructive, central and mixed apneas and hypopneas scored — so it tests the staging head *without* a montage confound and provides the only external test of the respiratory head. Because Sleep-EDF carries no cardiorespiratory channels, that branch is fed zeros there. Zeroing an input the network trained on is a train/test mismatch rather than an ablation, so we also trained a model with the branch zeroed throughout: it reaches 0.774 accuracy, below the 0.794 reported, which shows the headline figure is not an artefact of the mismatch. Both corpora also differ from iSLEEPS in montage, so both transfer figures are lower bounds rather than clean estimates. ISRUC records each subject twice; respiratory-event prevalence is 10.1% on the first night and 2.4% on the second, so we report the two separately rather than pooling a difference that has nothing to do with the model.

The respiratory read-out transfers. [STALE – every ISRUC number in this subsection is from the submitted model. ISRUC has not been re-run: the official archive has been unreachable since 5 Sep 2026, and the mirror used instead must pass the epoch-count check in the repository data notes before its numbers are trusted.] On ISRUC night one the model reaches 0.721 AUC against 0.782 in-domain — and it does so handicapped, because ISRUC provides no pulse channel and no separate effort trace, leaving two of the seven cardiorespiratory inputs zero-filled. A detector that had latched onto one centre's equipment would not survive that; this one does.

Staging moves in opposite directions on the two corpora, for reasons the data identify. On Sleep-EDF the model *gains* 0.083 of κ over its in-domain score. A representation that had memorised corpus-specific artefacts would degrade out of corpus, so the straightforward reading is that the features are physiological and that the ceiling reported throughout this paper is a property of *stroke sleep* rather than of the model — the same conclusion the learning curve reaches (Section VI-K) by a different route. On ISRUC staging falls by 0.059 accuracy, and the class mix explains much of it: ISRUC carries 15.6% N1 against iSLEEPS' 10.2%, and N1 is the weakest stage in every cohort we have measured (recall 0.201 on ISRUC, 0.247 on Sleep-EDF, and the lowest per-class F1 in-domain). [The N1 claim survives — N1 is still the weakest stage everywhere, at F1 0.306 on Sleep-EDF and 0.300 in-domain — but these two recall figures are from the submitted model and need recomputing.]

Comparing the two also isolates the montage effect. [STALE, AND THE CLAIM MAY HAVE FLIPPED. Under the final model Sleep-EDF N3 reaches F1 0.802, which is not consistent with a recall collapse to 0.554 — the argument in this paragraph may no longer hold. The external runner saves per-class F1 but not the confusion matrix, so N3 recall and the N3-read-as-N2 fraction have to be recomputed before this paragraph is either rewritten or removed. The ISRUC half additionally needs the corpus itself.] On Sleep-EDF, N3 recall collapses to 0.554 with 44% of N3 epochs read as N2, which we attributed to slow-wave activity being frontally dominant while the model was trained on central and occipital derivations. On ISRUC, where those derivations match ours exactly, **N3 recall recovers to** 0.654 while every other stage transfers worse than on Sleep-EDF. The prediction was made on one corpus and held on another chosen for an unrelated reason.

Two limits keep this in proportion. Accuracy across cohorts with different class balance is not directly comparable, which is why κ is reported alongside. And ISRUC contributes only eight subjects: per-recording accuracy spans 0.458–0.748 and per-recording respiratory AUC spans 0.515–0.848, the latter tracking event prevalence closely, so the low-prevalence nights give near-chance estimates.

P. MODEL COMPLEXITY AND CAPABILITY

Table 12 sets the proposed model beside representative alternatives on size and capability. A single compact network, under a million parameters and an order of magnitude smaller than the raw-signal transformers, delivers both a competitive stage and a respiratory read-out from one pass over the recording. None of the staging-only models, at any size, can supply the second output.

TABLE 11. External validation on two public corpora, zero-shot, one model trained on all 99 iSLEEPS patients with no dataset-specific tuning. In-domain figures are the ten-fold patient-independent means. Sleep-EDF carries no cardiorespiratory channels, so the respiratory head is not scored there. The two ISRUC rows are carried over from the feature-only configuration and have not been recomputed for the model reported here.

Corpus	Cohort	Rec.	Acc↑	κ↑	Resp. A
iSLEEPS (in-domain)	stroke	99	0.739	0.634	0.7
Sleep-EDF	healthy	40	**0.794**	**0.717**	–
ISRUC, night 1	sleep-disord.	8	0.664	0.557	**0.7**
ISRUC, night 2	sleep-disord.	8	0.597	0.454	0.7

TABLE 12. Model size and clinical capability. The proposed model is compact and is the only one that produces both a stage and a respiratory-event label.

Model	Input	Params↓	Outputs
CNN-ResNet18 [13]	1 EEG	∼11 M	stage
DeepSleepNet [2]	1 EEG	∼21 M	stage
Feature-sequence BiLSTM	7 ch feat	1.2 M	stage
Raw multimodal CNN	14 ch	0.95 M	stage + apnea
MM-Net (proposed)	14 ch feat	**0.86 M**	**stage + apnea**

VII. DISCUSSION

The result is a single model that reads the whole sleeping body and speaks to two clinical questions at once, in the population where both matter. The learned per-epoch embedding (Fig. 10) separates cleanly by sleep stage while the respiratory structure stays diffuse, mirroring the strong staging and the modest respiratory AUC.

Staging accuracy on this cohort saturates below 0.75 for every model family we evaluated, and the proposed model reaches 0.739 while adding a respiratory capability the staging models lack. The respiratory read-out does not depend on the labeling channel: it survives deletion of the airflow channel and rests on oxygen desaturation, effort, and arousal, the same signs a physician reads at the bedside.

The 0.72 staging figure should be judged on this cohort, where healthy-sleep accuracy does not transfer. Deep architectures developed on healthy sleepers collapse here: DeepSleepNet reaches 0.615, CNN-ResNet18 0.617, and a network pretrained on the healthy Sleep-EDF corpus and applied to this cohort 0.623, each roughly fifteen to twenty points below its healthy accuracy of about 0.85. The injured brain distorts the very micro-events, spindles, slow waves, and atonia, that those models learned to read. What holds up is physiologically grounded modeling on engineered features, at 0.72 to 0.75, which matches the published baseline on this corpus. The lesson is that healthy-sleep deep learning fails on the injured brain, and this cohort requires cohort-specific, physiologically grounded, multimodal modeling.

The study also draws a clean physiological line. The cortical channels carry sleep staging, and the cardiorespiratory channels carry breathing, and the model places each where it belongs rather than blurring them. That the neural and multimodal variants stage alike is the expected outcome: the

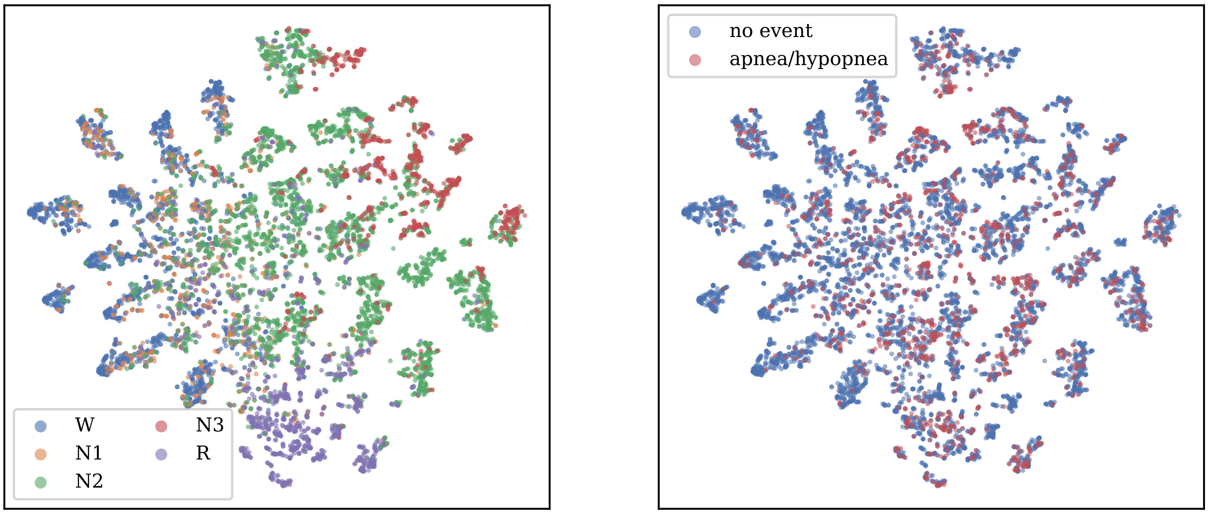

FIGURE 10. **Two-dimensional t-SNE of the model's per-epoch embeddings (pooled test epochs). Left: coloured by sleep stage – Wake, N2, N3, and REM form distinct territories while N1 is diffuse. Right: coloured by respiratory-event label – events are spread across the manifold, consistent with the modest detection AUC.**

second modality adds a new capability while leaving staging unchanged. We view this separation as a strength for clinical use, because a reader can trust that the respiratory output reflects breathing and the staging output reflects the brain.

A. LIMITATIONS

The evidence comes from a single center and the 96 patients with complete cardiorespiratory channels. Section VI-O shows both heads transfer to independent corpora without tuning, but neither external corpus is a stroke cohort, so every stroke-specific claim rests on iSLEEPS alone; multi-center validation on a second stroke cohort is required before clinical use, and no such public corpus currently exists. The external respiratory result rests on eight subjects, two of whose seven cardiorespiratory inputs were unavailable, so it establishes that the read-out transfers rather than how well. Respiratory detection, at 0.782 AUC, is a useful signal but not yet a screening-grade one, and it is scored at the epoch level rather than per event — at event level it detects 55% of events at 2.6 false alarms per hour and under-counts the event rate by more than half, so it cannot substitute for a scored index. The staging head is also overconfident (expected calibration error 0.076), so its posteriors should be used for ranking and abstention rather than read as probabilities. Staging sits at the cohort ceiling, so the model does not raise staging accuracy above the published baseline for this corpus. Finally, the model is feature-based: it inherits the assumptions of the engineered representations and is not an end-to-end raw-signal learner, a deliberate trade for data efficiency on a small cohort.

B. FUTURE WORK

Two directions follow naturally. Event-level metrics, such as sensitivity per event and false alarms per hour, would sit beside the epoch-level AUC and speak more directly to a sleep technician, and finer modeling of the breathing waveforms is likely to raise the respiratory number further. Linking the multimodal representation to stroke severity and recovery is the larger opportunity, and one the cardiorespiratory signal is well placed to serve.

VIII. CONCLUSION

We built a model that reads the full polysomnogram of stroke patients and, from one pass, stages their sleep and detects their respiratory events. Deep networks built for healthy sleep collapse to 0.61–0.69 accuracy on this cohort, so its 0.739 staging accuracy is at the cohort ceiling for every model family. The contribution is the second output, a respiratory read-out at 0.782 AUC that survives removal of the airflow channel, showing it reads the underlying physiology. Reading the whole recording gives the model a second clinical voice in the population that needs it most. Code and features are released.

DATA AND CODE AVAILABILITY

The iSLEEPS corpus is publicly available (Zenodo, DOI 10.5281/zenodo.14873844; and the india-data.org portal) as described by Maiti *et al.* [13]. Our code, engineered features, and reproduction notebook are released at https://github.com/wshuv-o/isleeps-sleep-staging.

ETHICS

The iSLEEPS data collection was approved by the NIMHANS Institutional Ethics Committee (No. NIMHANS/34th IEC (BS&NS DIV.)/2022, dated 05.02.2022) [13]. This work is a secondary analysis of the de-identified public release and required no additional enrollment.

TABLE 13. Recent single-channel sleep-staging methods: the accuracy each reports on healthy or general cohorts, set against its accuracy on the subacute-stroke cohort (iSLEEPS) for the two architectures we re-implemented. A dash marks a value we did not evaluate.

Method	Year	Approach	Reported acc (dataset)	On iSLEEPS
DeepSleepNet [2]	2017	Two representation-learning CNNs followed by a bidirectional LSTM sequence model, single-channel EEG.	0.82 (Sleep-EDF)	0.615
AttnSleep [5]	2021	Multi-resolution CNN with multi-head self-attention over epochs, single-channel EEG.	0.844 (Sleep-EDF)	0.686
SleepTransformer	2022	Transformer encoder with epoch-level interpretability and uncertainty estimates.	0.877 (SHHS)	–
1D-ResNet-SE-LSTM	2023	Residual network with squeeze-and-excitation blocks and an LSTM on raw EEG.	0.864 (Sleep-EDF)	–
Micro SleepNet	2023	Lightweight convolutional model for real-time mobile staging.	0.833 (SHHS)	–
AT-BiLSTM	2024	Attention-augmented bidirectional LSTM, single-channel EEG.	0.838 (Sleep-EDF)	–
MM-Net (proposed)	–	Two feature streams (neural and cardiorespiratory) fused into a BiLSTM that jointly stages sleep and detects respiratory events from the full polysomnogram.	–	**0.739**

FUNDING

The authors received no specific funding for this work.

ACKNOWLEDGMENT

The authors thank the iSLEEPS investigators for releasing the corpus.

REFERENCES

[1] American Academy of Sleep Medicine, *The AASM Manual for the Scoring of Sleep and Associated Events: Rules, Terminology and Technical Specifications, Version 3*. American Academy of Sleep Medicine, 2023.

[2] A. Supratak, H. Dong, C. Wu, and Y. Guo, "DeepSleepNet: A model for automatic sleep stage scoring based on raw single-channel EEG," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 25, no. 11, pp. 1998–2008, 2017.

[3] H. Phan, F. Andreotti, N. Cooray, O. Y. Chén, and M. De Vos, "SeqSleepNet: End-to-end hierarchical recurrent neural network for sequence-to-sequence automatic sleep staging," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 27, no. 3, pp. 400–410, 2019.

[4] A. Supratak and Y. Guo, "TinySleepNet: An efficient deep learning model for sleep stage scoring based on raw single-channel EEG," in *42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, 2020, pp. 641–644.

[5] E. Eldele, Z. Chen, C. Liu, M. Wu, C.-K. Kwoh, X. Li, and C. Guan, "An attention-based deep learning approach for sleep stage classification with single-channel EEG," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 29, pp. 809–818, 2021.

[6] H. Phan, O. Y. Chén, M. C. Tran, P. Koch, A. Mertins, and M. De Vos, "XSleepNet: Multi-view sequential model for automatic sleep staging," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 9, pp. 5903–5915, 2022.

[7] M. Perslev, M. H. Jensen, S. Darkner, P. J. Jennum, and C. Igel, "U-Time: A fully convolutional network for time series segmentation applied to sleep staging," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, pp. 4417–4428.

[8] W.-B. Jiang, L.-M. Zhao, and B.-L. Lu, "Large brain model for learning generic representations with tremendous EEG data in BCI," in *International Conference on Learning Representations (ICLR)*, 2024, arXiv:2405.18765.

[9] J. Wang, S. Zhao, Z. Luo, Y. Zhou, H. Jiang, S. Li, T. Li, and G. Pan, "CBraMod: A criss-cross brain foundation model for EEG decoding," in *International Conference on Learning Representations (ICLR)*, 2025, arXiv:2412.07236.

[10] G. Wang, W. Liu, Y. He, C. Xu, L. Ma, and H. Li, "EEGPT: Pretrained transformer for universal and reliable representation of EEG signals," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 37, 2024.

[11] C. Yang, M. B. Westover, and J. Sun, "BIOT: Biosignal transformer for cross-data learning in the wild," in *Advances in Neural Information Processing Systems 36 (NeurIPS)*, 2023, pp. 78 240–78 260. [Online]. Available: https://openreview.net/forum?id=c2LZyTyddi

[12] H. Phan and K. Mikkelsen, "Automatic sleep staging of EEG signals: Recent development, challenges, and future directions," *Physiological Measurement*, vol. 43, no. 4, 2022.

[13] S. Maiti, S. K. Sharma, S. Mythirayee, S. Rajendran, and R. S. Bapi, "Polysomnography dataset for sleep analysis in ischemic stroke patients," *Scientific Data*, vol. 13, no. 1, p. 421, 2026.

[14] C. L. Bassetti and M. S. Aldrich, "Sleep electroencephalogram changes in acute hemispheric stroke," *Sleep Medicine*, vol. 2, no. 3, pp. 185–194, 2001.

[15] Denis *et al.*, "Functional recovery after ischemic stroke: Impact of different sleep health parameters," *Journal of Sleep Research*, 2024.

[16] S. Bose, S. Maiti, S. K. Sharma, S. Mythirayee, T. Chakraborti, S. Rajendran, and R. S. Bapi, "AI generalisation gap in comorbid sleep disorder staging," in *IEEE International Symposium on Biomedical Imaging (ISBI)*, 2026.

[17] H. Phan, K. Mikkelsen, O. Y. Chén, P. Koch, A. Mertins, and M. De Vos, "SleepTransformer: Automatic sleep staging with interpretability and uncertainty quantification," *IEEE Transactions on Biomedical Engineering*, vol. 69, no. 8, pp. 2456–2467, 2022.

[18] H. Phan, K. P. Lorenzen, E. Heremans, O. Y. Chén, M. C. Tran, P. Koch, A. Mertins, M. Baumert *et al.*, "L-SeqSleepNet: Whole-cycle long sequence modelling for automatic sleep staging," *IEEE Journal of Biomedical and Health Informatics*, vol. 27, no. 10, pp. 4748–4757, 2023.

[19] N. Jirakittayakorn, Y. Wongsawat, and S. Mitrirattanakul, "ZleepAnlystNet: A novel deep learning model for automatic sleep stage scoring based on single-channel raw EEG data using separating training," *Scientific Reports*, vol. 14, no. 1, p. 9859, 2024.

[20] Y. Guo, M. Nowakowski, and W. Dai, "FlexSleepTransformer: A transformer-based sleep staging model with flexible input channel configurations," *Scientific Reports*, vol. 14, no. 1, p. 26312, 2024.

[21] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," *arXiv preprint arXiv:2312.00752*, 2023.

[22] X. Zhou, Y. Han, Z. Chen, C. Liu, Y. Ding, Z. Jia, and Y. Liu, "BiT-MamSleep: Bidirectional temporal mamba for EEG sleep staging," *arXiv preprint arXiv:2411.01589*, 2024.

[23] M. Perslev, S. Darkner, L. Kempfner, M. Nikolic, P. J. Jennum, and C. Igel, "U-Sleep: Resilient high-frequency sleep staging," *npj Digital Medicine*, vol. 4, no. 1, p. 72, 2021.

[24] R. Thapa, M. R. Kjaer, B. He, I. Covert, H. Moore IV, U. Hanif *et al.*, "A multimodal sleep foundation model for disease prediction," *Nature Medicine*, 2026.

[25] Y. Zhou, J. Wu, Z. Ren, Z. Yao, W. Lu, K. Peng, Q. Zheng, C. Song *et al.*, "CSBrain: A cross-scale spatiotemporal brain foundation model for EEG decoding," *arXiv preprint arXiv:2506.23075*, 2025.

[26] K. Kontras, T. Osselaer, S. G. Mouslech, A.-I. Karaiskou, G. Gagliardi, T. Strypsteen *et al.*, "NeuroAtlas: Benchmarking foundation models for clinical EEG and brain-computer interfaces," *arXiv preprint arXiv:2605.14698*, 2026.

[27] A. Dei Rossi, M. Metaldi, M. Bechny, I. Filchenko, J. van der Meer, M. H. Schmidt, C. L. A. Bassetti, A. Tzovara *et al.*, "SLEEPYLAND: Trust begins with fair evaluation of automatic sleep staging models," *npj Digital Medicine*, vol. 9, no. 1, p. 55, 2026.

[28] E. Estevan, M. Sierra-Torralba, E. López-Larraz, and L. Montesano, "A systematic evaluation of self-supervised learning for label-efficient sleep staging with wearable EEG," *arXiv preprint arXiv:2510.07960*, 2025.

[29] A. Guillot and V. Thorey, "RobustSleepNet: Transfer learning for automated sleep staging at scale," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 29, pp. 1441–1451, 2021.

[30] K. Yang and H. Wang, "Deep learning for EEG-based sleep stage classification: A review," *Medical & Biological Engineering & Computing*, vol. 64, pp. 2017–2043, 2026.

[31] U. Hanif, A. Aloulou, F. Crosbie, P. Bouchequet, M. Chennaoui, T. Andrillon, and D. Leger, "Deciphering insomnia: Benchmarking automated sleep staging algorithms for complex sleep disorders," *Journal of Sleep Research*, vol. 34, no. 6, p. e70048, 2025.

[32] T. Jing, L. Xu, S. Sun, W. Yu, Y. Zheng, G. Xu, X. Shen, T. Tang, and C. Chu, "CPAP modulation of sleep microstructure as predictors of neurological recovery in ischemic stroke patients with moderate-to-severe OSA," *Frontiers in Neurology*, vol. 16, p. 1605106, 2025.

[33] N. Tellenbach, M. H. Schmidt, F. Alexiev, E. Blondiaux, F. Cavalloni, C. L. Bassetti, L. Heydrich, and P. Bargiotas, "REM sleep and muscle atonia in brainstem stroke: A quantitative polysomnographic and lesion analysis study," *Journal of Sleep Research*, vol. 32, no. 2, 2022.

[34] J. M. Gottselig, C. L. Bassetti, and P. Achermann, "Power and coherence of sleep spindle frequency activity following hemispheric stroke," *Brain*, vol. 125, no. 2, pp. 373–383, 2002.

[35] L. M. J. Fernández and A. Lüthi, "Sleep spindles: Mechanisms and functions," *Physiological Reviews*, vol. 99, no. 2, pp. 1349–1385, 2019.

[36] B. K. Simpson, R. Rangwani, A. Abbasi, J. M. Chung, C. M. Reed, and T. Gulati, "Disturbed laterality of non-rapid eye movement sleep oscillations in post-stroke human sleep: A pilot study," *Frontiers in Neurology*, vol. 14, p. 1243575, 2023.

[37] L. Facchin, C. Schöne, A. Mensen *et al.*, "Slow waves promote sleep-dependent plasticity and functional recovery after stroke," *Journal of Neuroscience*, vol. 40, no. 45, pp. 8637–8651, 2020.

[38] Y. Zheng, T. Jing, L. Xu, S. Sun, W. Yu, R. Liu, G. Xu, and C. Chu, "Analysis of the correlation between sleep spindles and cognitive impairment in patients with ischemic stroke," *Scientific Reports*, vol. 15, no. 1, p. 36259, 2025.

[39] A. Agius Anastasi *et al.*, "Brain symmetry index in healthy and stroke patients for assessment and prognosis," *Stroke Research and Treatment*, vol. 2017, 2017.

[40] B. Kemp, A. H. Zwinderman, B. Tuk, H. A. C. Kamphuisen, and J. J. L. Oberyé, "Analysis of a sleep-dependent neuronal feedback loop: The slow-wave microcontinuity of the EEG," *IEEE Transactions on Biomedical Engineering*, vol. 47, no. 9, pp. 1185–1194, 2000.

[41] A. L. Goldberger, L. A. N. Amaral, L. Glass *et al.*, "PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals," *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000.

[42] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2016, pp. 785–794.

[43] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems 30 (NIPS)*, 2017, no DOI is registered for this volume.

[44] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: Unbiased boosting with categorical features," in *Advances in Neural Information Processing Systems 31 (NeurIPS)*, 2018.

[45] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in *International Conference on Learning Representations (ICLR)*, 2018.

[46] A. N. Angelopoulos and S. Bates, "Conformal prediction: A gentle introduction," *Foundations and Trends in Machine Learning*, vol. 16, no. 4, pp. 494–591, 2023.

[47] A. Lacoste, A. Luccioni, V. Schmidt, and T. Dandres, "Quantifying the carbon emissions of machine learning," *arXiv preprint arXiv:1910.09700*, 2019.

[48] B. Hjorth, "EEG analysis based on time domain properties," *Electroencephalography and Clinical Neurophysiology*, vol. 29, no. 3, pp. 306–310, 1970.

[49] W. Chen, Y. Cai, A. Li, Y. Su, and K. Jiang, "Single-channel sleep EEG classification method based on LSTM and hidden markov model," *Brain Sciences*, vol. 14, no. 11, p. 1087, 2024.

[50] S. Khalighi, T. Sousa, J. M. Santos, and U. Nunes, "ISRUC-Sleep: A comprehensive public dataset for sleep researchers," *Computer Methods and Programs in Biomedicine*, vol. 124, pp. 180–192, 2016.

MD WAHIDUZZAMAN SUVA is with the Department of Computer Science and Engineering, American International University-Bangladesh (AIUB), Dhaka, Bangladesh. His research interests include deep learning for biomedical signal processing, sleep staging from polysomnography, and multimodal representation learning. [degree(s), year(s) and awarding institution to be added]

ESM-E MOULA CHOWDHURY ABHA is with the Department of Computer Science and Engineering, American International University-Bangladesh (AIUB), Dhaka, Bangladesh. Her research interests include machine learning for health informatics, physiological time-series analysis, and multimodal deep learning. [degree(s), year(s) and awarding institution to be added; confirm pronoun]

MD IMTIAJ ALAM SAJIN is with the Department of Computer Science and Engineering, American International University-Bangladesh (AIUB), Dhaka, Bangladesh. His research interests include deep learning, biomedical signal analysis, and applied machine learning. [degree(s), year(s) and awarding institution to be added]

MD IFTEKHARUL MOBIN is an Assistant Professor with the Department of Computer Science, American International University-Bangladesh (AIUB), Dhaka, Bangladesh, where he holds a doctoral degree. His research interests include data analysis, machine learning, natural language processing, and data mining. He is the corresponding author of this work. [Ph.D. awarding institution and year, prior appointments, and any professional memberships to be added – the AIUB faculty page carries no populated detail and these could not be verified.]